TOP 50 REPEATING CHEMISTRY MCQS

Revision Series | Made By Pragya (Thunderstudy)

TRANSITION ELEMENTS & COORDINATION

1. Which first row transition element has highest third ionisation enthalpy?

A. Fe
B. Mn
C. Cr
D. V

✅ **Answer: B (Mn)**

2. Which transition metal ion is colourless?

A. Cu^+
B. Ni^{2+}
C. Co^{2+}
D. V^{3+}

✅ **Answer: A (Cu^+)**

3. Transition metals have high melting points due to:

A. Strong metallic bonding
B. Low density
C. Weak nuclear charge
D. High reactivity

✅ **Answer: A**

4. Which complex is diamagnetic?

A. $[Cr(NH_3)_6]^{3+}$
B. $[Ni(CN)_4]^{2-}$
C. $[Fe(H_2O)_6]^{3+}$
D. $[Ti(H_2O)_6]^{3+}$

✅ **Answer: B**

5. Zr and Hf have similar atomic radii due to:

A. Shielding effect
B. Lanthanoid contraction
C. Ionisation energy
D. Hybridisation

✅ **Answer: B**

ORGANIC CHEMISTRY

6. Fastest SN1 reaction occurs in:

A. Primary halide
B. Secondary halide
C. Tertiary halide
D. Methyl halide

✅ **Answer: C**

7. Acetic acid reacts with PCl_5 to form:

A. CH_3COCl
B. CH_3COOH
C. $CH_2ClCOCl$
D. CCl_3COOH

✅ **Answer: A**

8. Cyanohydrin formation is example of:

A. Nucleophilic addition
B. Electrophilic addition
C. Nucleophilic substitution
D. Electrophilic substitution

✅ **Answer: A**

9. Alcohol that does NOT undergo oxidation:

A. Primary
B. Secondary
C. Tertiary
D. Allylic

✅ **Answer: C**

10. Ethyl bromide + alcoholic $AgNO_2$ gives:

A. Nitroethane
B. Ethyl nitrite
C. Ethene
D. Methane

✅ **Answer: A**

11. $CH_3CH_2OH \rightarrow CH_3CHO$ uses:

A. PCC
B. $KMnO_4$
C. H_2/Pd
D. $LiAlH_4$

✅ **Answer: A**

12. Compound giving ketone on oxidation with CrO_3:

A. Primary alcohol
B. Secondary alcohol
C. Tertiary alcohol
D. Phenol

✅ **Answer: B**

13. Auto-oxidation of chloroform forms:

A. Phosgene
B. Mustard gas
C. Tear gas
D. Ammonia

✅ **Answer: A**

14. Sandmeyer reaction converts:

A. Diazonium salt → Aryl halide
B. Alcohol → Alkane
C. Alkene → Alcohol
D. Amide → Amine

✅ **Answer: A**

15. Strongest base among given species depends on:

A. Resonance stability
B. Molecular weight
C. Hybridisation
D. Hydrogen bonding

✅ **Answer: A**

CHEMICAL KINETICS

16. Catalyst changes:

A. Activation energy
B. Enthalpy
C. Equilibrium constant
D. Gibbs energy

✅ **Answer: A**

17. Slope of log k vs 1/T graph equals:

A. Ea/2.303R
B. 2.303R/Ea
C. A
D. RT/Ea

✅ **Answer: A**

18. Rate of zero order reaction depends on:

A. Concentration
B. Temperature
C. Independent of concentration
D. Pressure

✅ **Answer: C**

19. Rate increases 16 times when concentration increases 4 times. Order =

A. 1
B. 2
C. 3
D. 0

✅ **Answer: B**

20. For X + 2Y → P, correct relation:

A. dP/dt = – dX/dt
B. dP/dt = –½ dY/dt
C. dX/dt = dY/dt
D. None

✅ **Answer: A**

SOLUTIONS & COLLIGATIVE PROPERTIES

21. Highest freezing point solution:

A. 1M KCl
B. 1M Na_2SO_4
C. 1M Glucose
D. 1M $AlCl_3$

✅ **Answer: C**

22. Association of electrolyte gives Van't Hoff factor:

A. >1
B. <1
C. =1
D. 0

✅ **Answer: B**

23. Boiling point elevation depends on:

A. Molality
B. Molarity
C. Pressure
D. Volume

✅ **Answer: A**

24. Isotonic solutions have same:

A. Density
B. Osmotic pressure
C. Volume
D. Refractive index

✅ **Answer: B**

25. On dilution:

A. Conductivity increases
B. Molar conductivity increases
C. Both decrease
D. Both remain constant

✅ **Answer: B**

ELECTROCHEMISTRY

26. Electrolysis of dilute H_2SO_4 gives gas at anode:

A. H_2
B. O_2
C. SO_2
D. SO_3

✅ **Answer: B**

27. Conductivity mainly depends on:

A. Number of ions
B. Temperature
C. Pressure
D. Colour

✅ **Answer: A**

28. Standard electrode potential indicates:

A. Oxidising strength
B. Atomic size
C. Density
D. Magnetic property

✅ **Answer: A**

BIOMOLECULES & POLYMERS

29. DNA contains:

A. Deoxyribose + Base + Phosphoric acid
B. Ribose + Base + Sulphuric acid
C. Deoxyribose + Uracil
D. Ribose + Adenine

✅ **Answer: A**

30. Protein primary structure means:

A. Sequence of amino acids
B. Folding pattern
C. Hydrogen bonding
D. Cross linkage

✅ **Answer: A**

HALOALKANES & HALOARENES

31. SN1 reaction favoured by:

A. Polar protic solvent
B. Polar aprotic solvent
C. Strong nucleophile
D. Low temperature

✅ **Answer: A**

32. Allyl halide high reactivity due to:

A. Resonance stabilisation
B. Inductive effect
C. Hybridisation
D. Hydrogen bonding

✅ **Answer: A**

AMINES & DIAZONIUM

33. Alkyl amines stronger base than ammonia due to:

A. +I effect
B. –I effect
C. Resonance
D. Hybridisation

✅ **Answer: A**

34. Aryldiazonium salts produce phenol by:

A. Hydrolysis
B. Reduction
C. Oxidation
D. Addition

✅ **Answer: A**

ALCOHOLS, PHENOLS & ETHERS

35. Dehydration of alcohol gives:

A. Alkene
B. Ketone
C. Ether
D. Aldehyde

✅ **Answer: A**

36. Phenol acidity increases due to:

A. Resonance stabilisation

B. Hydrogen bonding

C. Hyperconjugation

D. Inductive donation

✅ **Answer: A**

BASIC CONCEPTS & MISCELLANEOUS

37. Strongest intermolecular force in alcohols:

A. Hydrogen bonding

B. Dipole forces

C. London forces

D. Ionic bond

✅ **Answer: A**

38. Chiral carbon must have:

A. Four different groups

B. Double bond

C. Lone pair

D. Aromatic ring

✅ **Answer: A**

39. Fuel cell used in Apollo mission:

A. H_2–O_2 cell

B. Dry cell

C. Mercury cell

D. Ni–Cd cell

✅ **Answer: A**

40. Unit of rate constant indicates:

A. Reaction order

B. Molecularity

C. Temperature

D. Pressure

✅ **Answer: A**

41. Strong oxidising lanthanoid ion:

A. Ce^{4+}

B. Ce^{3+}

C. La^{3+}

D. Nd^{3+}

✅ **Answer: A**

42. Crystal field theory explains:

A. Colour of complexes
B. Molecular weight
C. Density
D. Solubility

✅ **Answer: A**

43. Highest boiling point solution has:

A. Highest Van't Hoff factor
B. Lowest molar mass
C. Lowest concentration
D. Highest temperature

✅ **Answer: A**

44. Electrophile is species that:

A. Accepts electron pair
B. Donates electron pair
C. Accepts proton
D. Donates neutron

✅ **Answer: A**

45. Nucleophile is species that:

A. Donates electron pair
B. Accepts electron pair
C. Accepts proton
D. Donates neutron

✅ **Answer: A**

46. Grignard reagent reacts with carbonyl to give:

A. Alcohol
B. Alkane
C. Ester
D. Ether

✅ **Answer: A**

47. Tollens reagent detects:

A. Aldehyde
B. Ketone
C. Alcohol
D. Amine

✅ **Answer: A**

48. Iodoform test indicates:

A. Methyl ketone
B. Ether
C. Alkene
D. Ester

✅ **Answer: A**

49. Highest boiling among isomers due to:

A. Hydrogen bonding
B. Steric effect
C. Branching
D. Polarity decrease

✅ **Answer: A**

50. Alcohol dehydration order:

A. 3° > 2° > 1°
B. 1° > 2° > 3°
C. 2° > 1° > 3°
D. All equal

✅ **Answer: A**

Practice these regularly for best results. Best of luck for your exams!
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